

Airbnb Market Performance, Sentiment And Pricing Intelligence

An End-to-End Data Analysis and
Interactive Dashboard

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Dataset Scope : 29,954 verified NYC properties (listings.csv), 11+ million forward booking records (calendar.csv), and 990,000+ customer reviews (reviews.csv)

Data Cleaning And Integration Pipeline

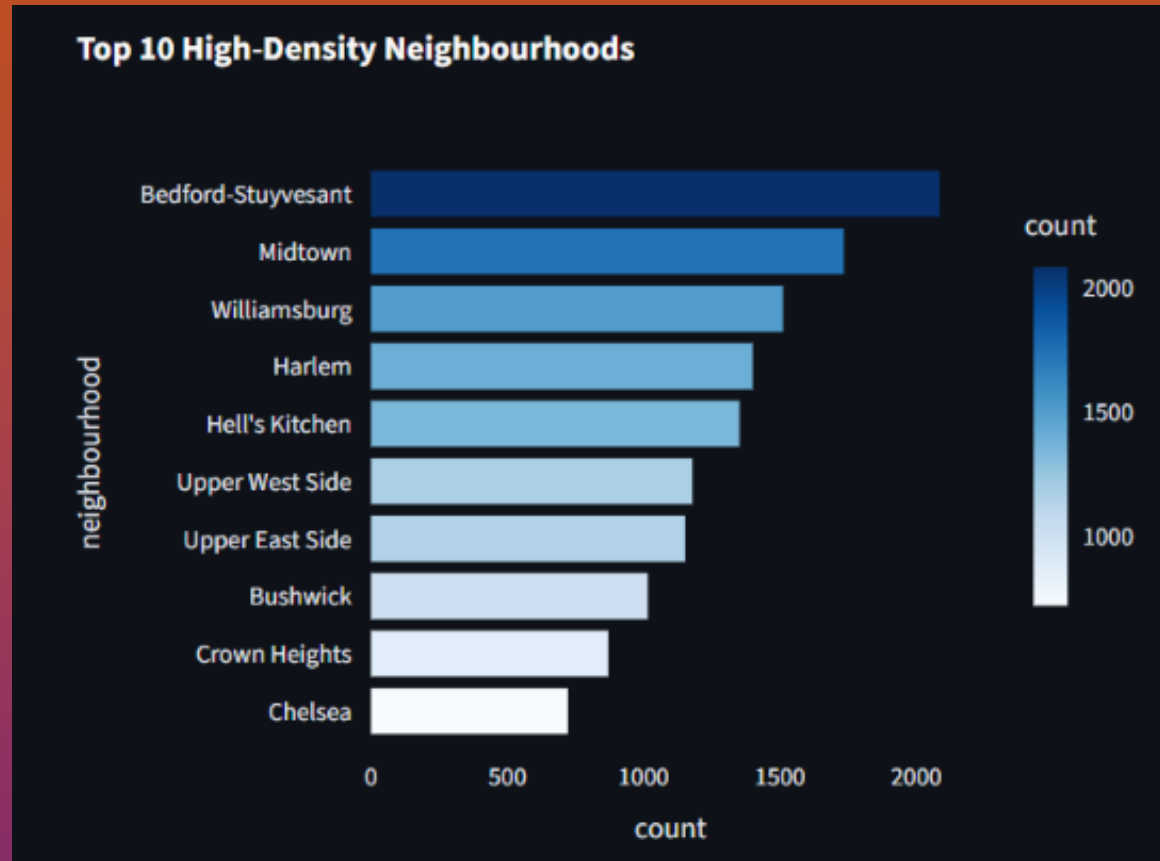
Data Cleansing: Handled currency symbols and non-numeric text in price, imputed missing prices using borough-room-type medians, and capped extreme 99th-percentile outliers (\$1,426 cap)

Cross-Dataset Merging:

- Integrated **NLP review sentiment polarity** (TextBlob) per property.
- Calculated true **forward occupancy rates** by aggregating booked (available == 'f') vs. open calendar nights.

Cleaned Base: A unified master dataset of 29,954 listings ready for analytics and dashboarding.

Geographic Distribution And Supply Density

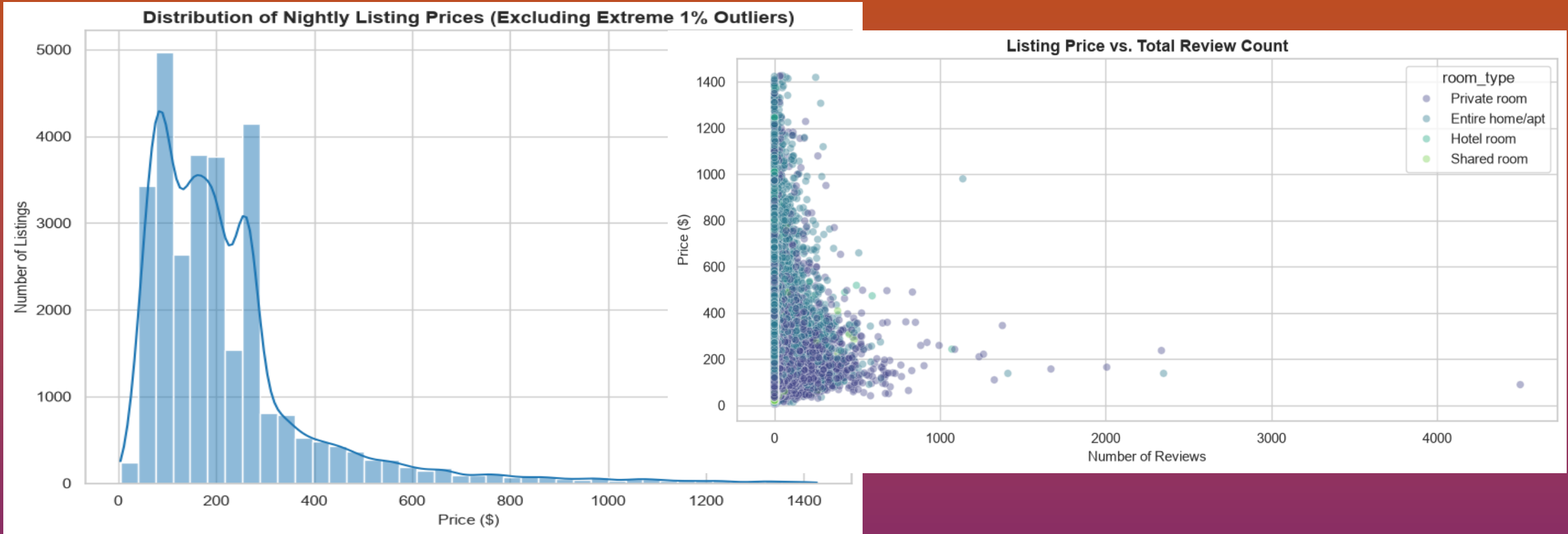


Key Findings:

- Heavy supply concentration in **Brooklyn and Manhattan**, representing over 80% of total NYC volume.
- Leading hubs: **Bedford-Stuyvesant (2,087 listings)**, **Midtown (1,735 listings)**, **Williamsburg (1,512 listings)**, and **Harlem (1,401 listings)**.

Market Insight: Outer boroughs offer volume and affordability, while central Manhattan commands premium pricing density.

Pricing Dynamics And Room-Type Analysis



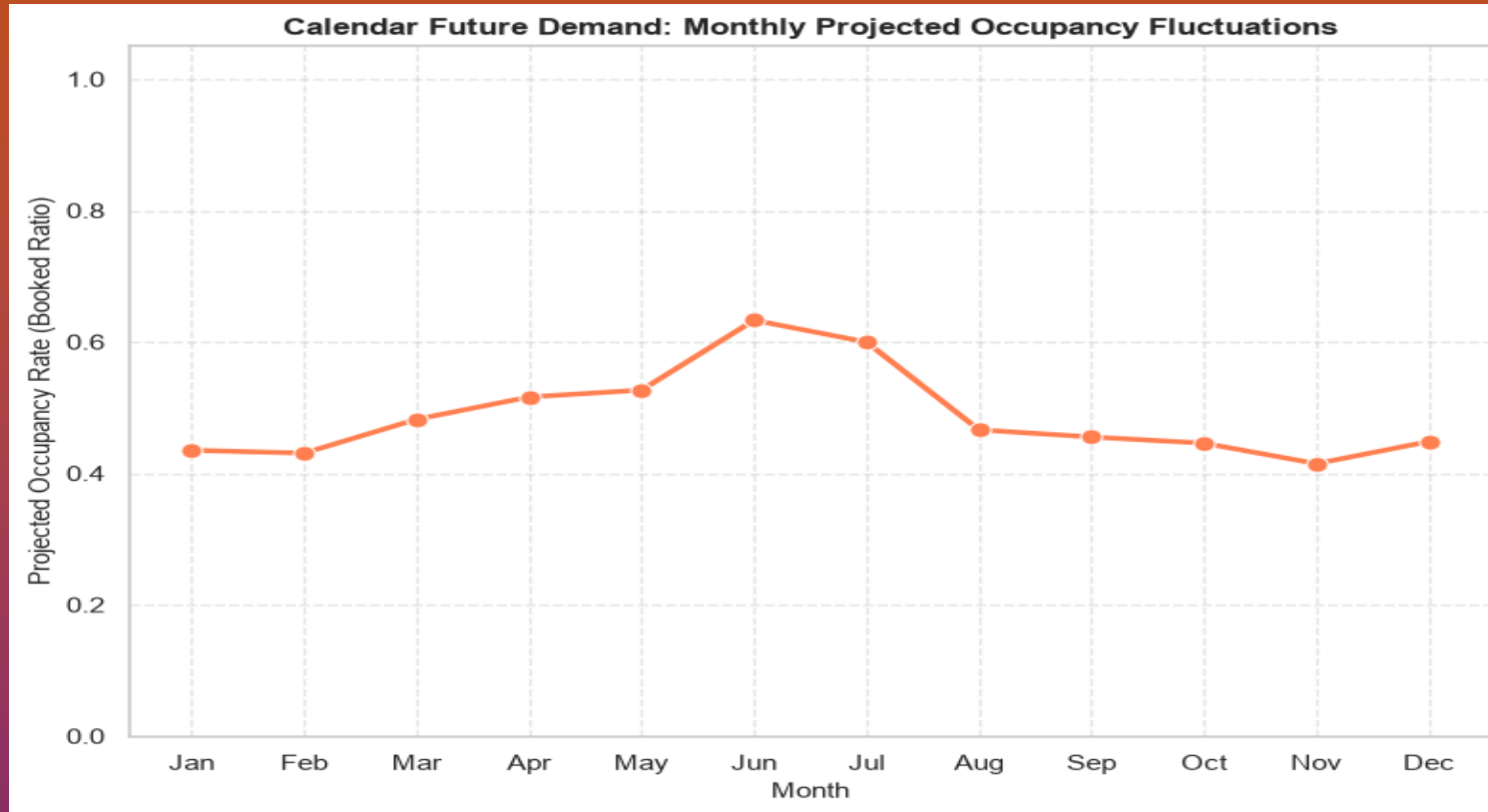
Key Findings:

Average nightly rate across all properties is around **\$200**.

Entire Homes/Apartments command a 2x–3x price premium over **Private Rooms**.

Scatter Plot Insight: High-review, frequent-turnover properties cluster between \$75 and \$180/night; ultra-high-priced listings see slower booking velocity

Calendar Demand And Seasonal Fluctuations



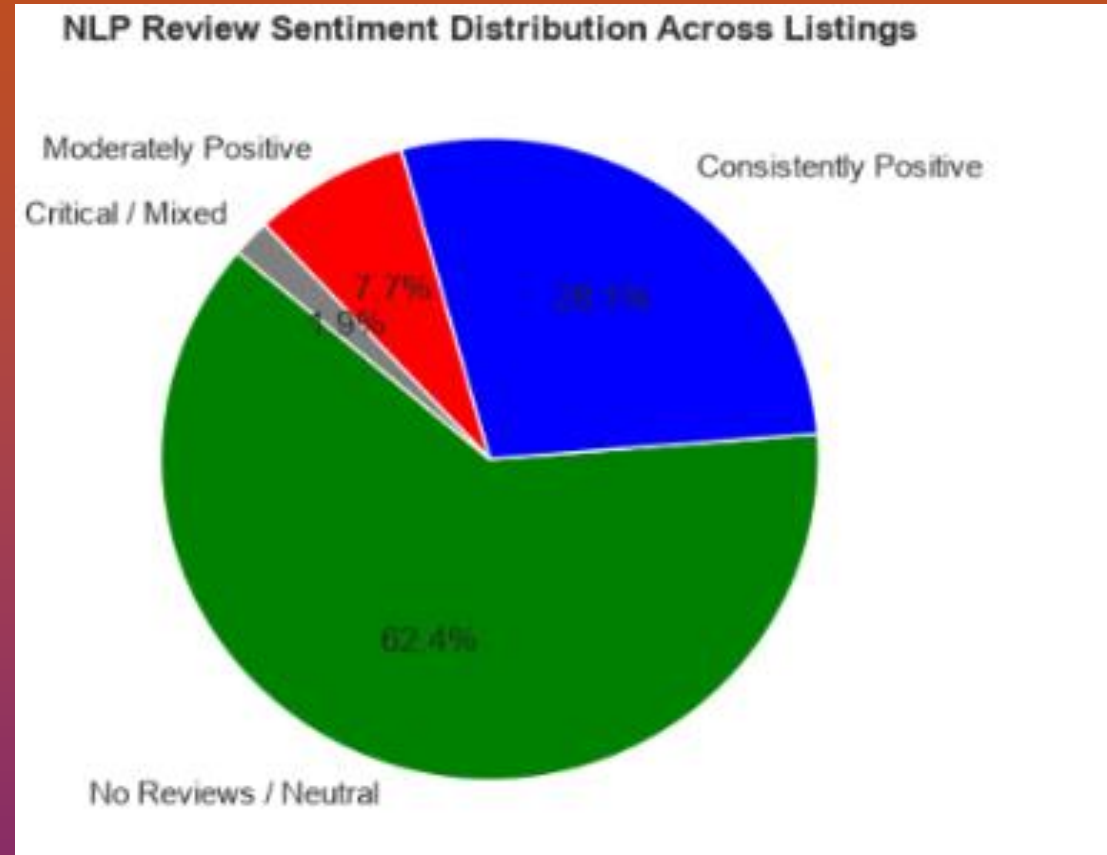
Key Findings:

Overall forward occupancy sits at **49.4%**.

Peak Season: Strong summer demand peaking in **June (63%)** and **July (60%)**.

Trough Period: Late autumn and winter experience vacancy dips, reaching lows in **November (41%)**, **February (43%)**, and **January (44%)**.

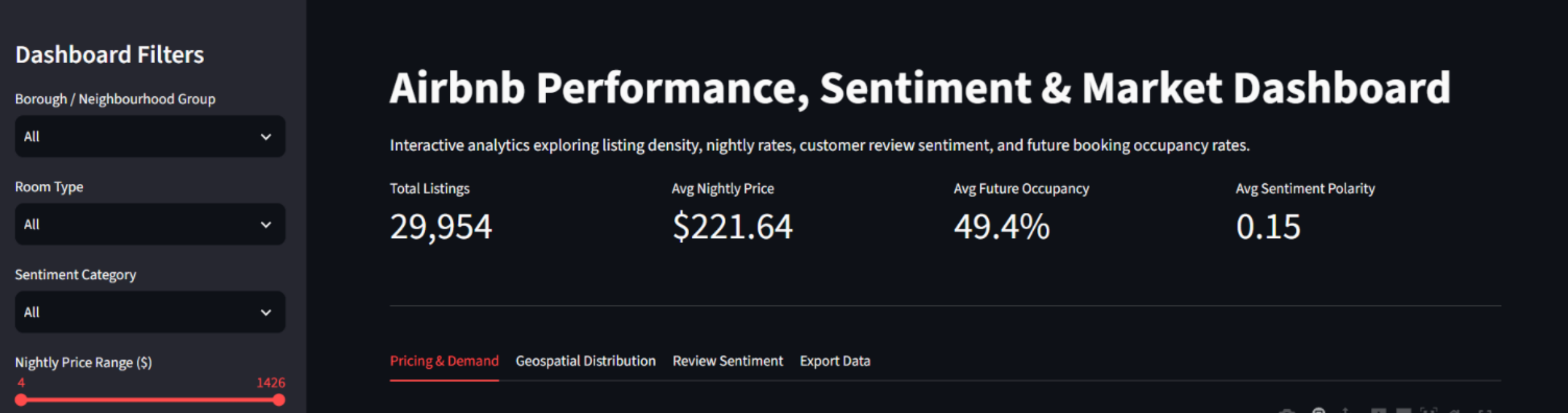
Customer Review Sentiment Analysis



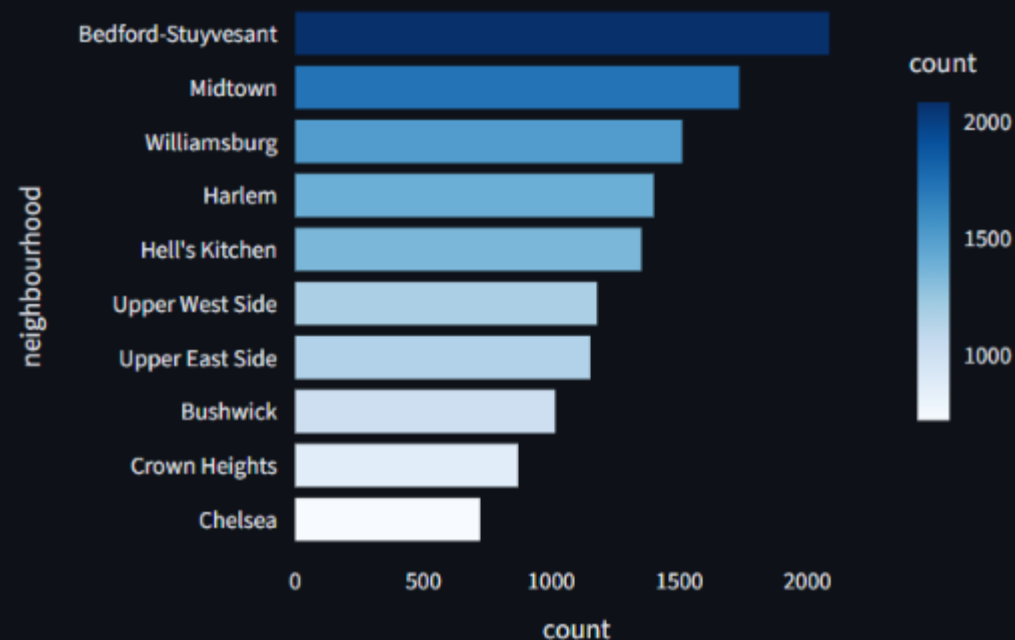
Key Findings:

- Mean guest sentiment polarity is **+0.15** across the board.
- Among reviewed listings, customer perception is overwhelmingly positive: **28.1% Consistently Positive** (8,417 properties) and **7.7% Moderately Positive**.
- Only **about 2.0%** exhibit Critical/Mixed reviews, predominantly driven by cleanliness and check-in friction.

Interactive Dashboard Architecture And Host Recommendations



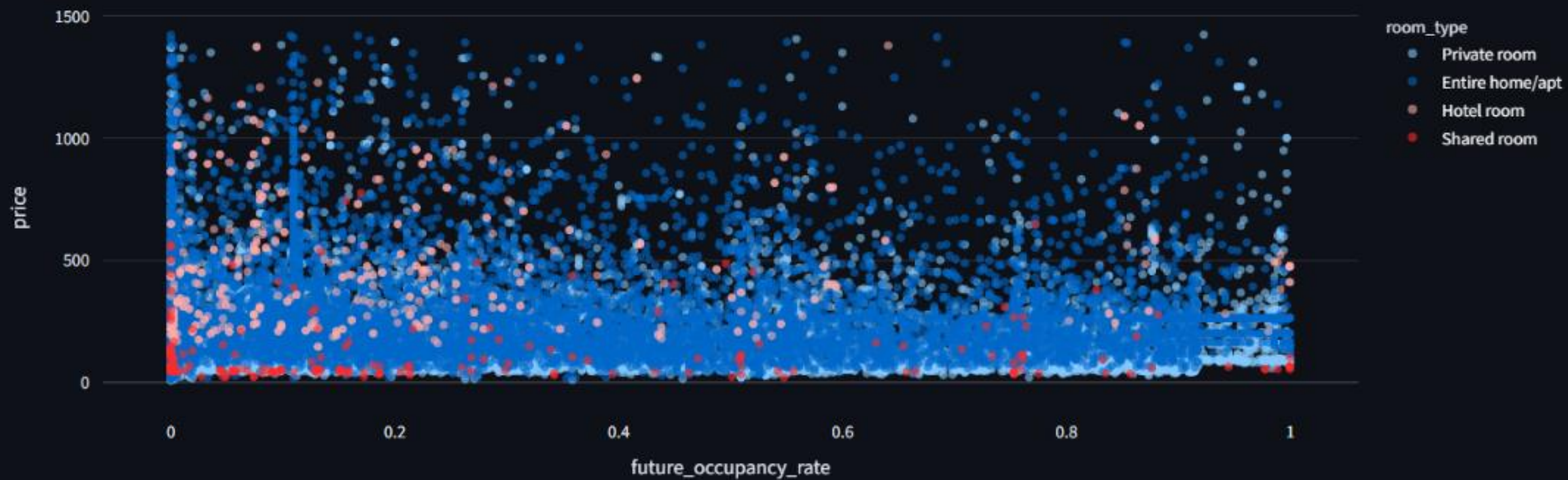
Top 10 High-Density Neighbourhoods



Nightly Price Spread by Room Type



Future Occupancy Rate vs. Nightly Price



Dashboard Features :

Slicers for Borough, Room Type, Sentiment Category, and Price Range.

Live KPI cards, Geospatial map, Plotly charts, and CSV data export.

Actionable Host Recommendations :

Dynamic Seasonal Pricing: Increase rates by 15%–20% during June/July peaks, while offering competitive weekly discounts in November–February to mitigate vacancy.

Convert Excess Space: In high-density boroughs, converting underutilized large units into private suites can increase bookings and review volume.

Address Review Velocity: Focus on check-in automation and cleanliness consistency to convert neutral listings into top-ranked positive properties.